Data Collection and Data Pre-Processing

Data Collection Strategies

- Intelligence Needs Data
- Human intelligence needs data:
- A real estate broker needs data about sold houses to estimate prices.
- Artificial Intelligence also needs data:
- A Machine Learning program needs data to estimate prices.
- Data can help us to see and understand.
- Data can help us to find new opportunities.
- Data can help us to resolve misunderstandings.

- **Data Collection Strategies**
- Data collection involves systematically gathering information from various sources. Common strategies include:
- **Primary Data Collection:** Gathering new, first-hand data for a specific purpose.
  - **Surveys & Questionnaires:** Efficient for reaching large audiences; tools like Google Forms or SurveyMonkey are commonly used.
  - **Interviews & Focus Groups:** Provide in-depth qualitative insights through direct interaction.
  - **Observations:** Recording behaviors in natural settings without direct interference.
  - **IoT & Sensor Data:** Capturing real-time telemetry (e.g., temperature, motion) from connected devices.
- **Secondary Data Collection:** Utilizing existing data collected by others.
  - **Public Datasets:** Accessing repositories like Kaggle or the UCI Machine Learning Repository.

Data Pre-Processing Overview

- Data preprocessing is the foundational process of converting raw, "dirty" data into a clean, structured, and high-quality format suitable for machine learning (ML) and artificial intelligence (AI). It is a critical stage that can consume up to **60% to 80%** of a data professional's time because the accuracy of any model is directly dependent on the quality of its input data.

Data Pre-Processing Overview

- **Core Tasks in Data Preprocessing**
- Modern preprocessing workflows typically involve these primary stages:
- **Data Cleaning:** Identifying and fixing errors to improve reliability.
  - **Handling Missing Values:** Techniques include **imputation** (filling gaps with mean, median, or mode) or **deletion** (removing incomplete rows).
  - **Noisy Data Removal:** Smoothing techniques like **binning**, regression, or clustering are used to eliminate outliers and irrelevant signals.
- **Data Integration:** Merging data from multiple sources (e.g., databases, flat files, or APIs) into a single, unified view.
- **Data Transformation:** Adjusting data to fit model requirements.
  - **Normalization & Scaling:** Rescaling numerical features (e.g., to a 0–1 range) so that features with larger magnitudes do not dominate the model.
  - **Encoding:** Converting categorical or text-based data (e.g., "Yes/No") into numerical formats like **One-Hot Encoding** or **Label Encoding**.
- **Data Reduction:** Reducing dataset size while preserving essential information to improve computational efficiency.

Data Cleaning

- **Data Cleaning** (also known as data cleansing or scrubbing) remains the most critical stage of data preprocessing. It is the process of identifying and fixing or removing errors, inconsistencies, and inaccuracies to ensure a dataset is reliable for analysis and AI modeling

- **Handling Missing Values:** Gaps caused by incomplete collection or entry errors.
  - **Imputation:** Filling gaps with statistical estimates like the mean, median, or mode.
  - **Advanced Imputation:** Using algorithms like **K-Nearest Neighbors (KNN)** or regression to predict missing data based on other variables.
  - **Deletion:** Removing entire rows or columns if the missingness is extensive and cannot be accurately imputed.
- **Removing Duplicates:** Eliminating identical or near-identical entries—often identified through **fuzzy matching**—that can skew results or over-represent certain data points.
- **Detecting and Managing Outliers:** Identifying extreme values that deviate significantly from the rest of the data using methods like **Z-score** (standard deviations from the mean) or the **IQR (Interquartile Range)** method.
- **Correcting Structural Errors:** Fixing typos, inconsistent capitalization (e.g., “RTN" vs. “Ratnagiri"), and mismatched naming conventions.
- **Standardizing Formats:** Ensuring uniformity in date formats (e.g., ISO 8601), units of measurement, and currency.

Data Integration and Transformation

- **1. Data Integration**
- Data integration involves combining data from multiple heterogeneous sources (e.g., databases, flat files, cloud APIs) into a single, unified view.
- **Standard Approaches:**
  - **Tight Coupling (Data Warehousing):** Data is physically extracted from sources, transformed, and stored in a centralized repository like a data warehouse.
  - **Loose Coupling (Data Federation):** Creates a virtual layer that allows users to query data from multiple sources in real time without physically moving it.
- **Modern Techniques in 2026:**
  - **Extract, Load, Transform (ELT):** Reverses traditional ETL by loading data into high-power cloud warehouses (like Snowflake or BigQuery) before transforming it, taking advantage of modern scalable compute.
  - **Change Data Capture (CDC):** Efficiently tracks and replicates only the *changes* in source systems (insertions, deletions, updates) to keep target systems synchronized in near real-time.
  - **Agentic Orchestration:** In 2026, AI agents autonomously plan and execute integration workflows, often using natural language prompts like "Sync Salesforce leads to Snowflake daily".

- **2. Data Transformation**
- Data transformation converts raw data into a format suitable for specific mining or modeling tasks.
- **Key Transformation Strategies:**
  - **Normalization:** Scaling numerical features to a standard small range (e.g., 0 to 1 or -1 to 1) to ensure all variables contribute equally to the model.
  - **Standardization (Z-score Scaling):** Adjusting data to have a mean of zero and a standard deviation of one, essential for algorithms that assume a normal distribution.
  - **Categorical Encoding:** Converting text-based categories into numbers. Common methods include **One-Hot Encoding** (creating binary columns) and **Label/Ordinal Encoding** (assigning unique integers).
  - **Aggregation:** Summarizing data (e.g., calculating monthly sales from daily records) to create more manageable and insightful datasets.
  - **Discretization (Binning):** Converting continuous numerical data into discrete intervals or "bins" (e.g., grouping ages into "Child," "Youth," or "Adult").
- **High-Quality Input:** Preprocessing is vital because the quality of input data directly
- **Efficiency:** Automated transformation reduces manual effort, **Consistency:** Integration eliminates data silos

Data Reduction

- **Data Reduction** is a vital preprocessing stage that aims to obtain a reduced representation of a dataset that is much smaller in volume yet maintains the integrity of the original data. This process is essential for managing the massive datasets common in modern big data and AI environments, as it significantly enhances computational efficiency and reduces storage costs

Data Discretization

- is a pivotal preprocessing technique used to convert continuous numerical data into discrete categories or "bins". By grouping infinite possible values into a finite set of intervals, discretization simplifies complex datasets, reduces noise, and makes data compatible with algorithms that require categorical inputs, such as **Decision Trees** and **Naïve Bayes**

Case Study Data Preprocessing in Credit Card Consuming Behavior Mining

